Varun Lochab

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Education

Columbia University, New York, USA
Ph.D., Physics (expected Spring 2027) — *Sep 2021 – Present*
Advisor: ***Frederik Denef***

Columbia University, New York, USA
M.A. & M.Phil., Physics, GPA: 4.30/4.33 — *May 2023 & May 2025*

The Ohio State University, Columbus, USA
M.S. & Ph.D., Mechanical Engineering — *Aug 2015 – Dec 2020*
Advisor: ***Shaurya Prakash***

Work Experience

Graduate Teaching & Research Assistant
Department of Physics, Columbia University, New York, USA — *Sep 2021 – Present*
GRA: Summer 2025, Summer 2026; GTA: all other terms.

Postdoctoral Researcher
Department of Physics
University of Wisconsin-Madison, Madison, USA — *Jan 2021 – Sep 2021*

Graduate Teaching & Research Assistant
Department of Mechanical & Aerospace Engineering
The Ohio State University, Columbus, USA — *Aug 2015 – Dec 2020*
GRA throughout; GTA: Spring 2020.

Publications & Preprints

Physics (†authors listed alphabetically)

1. **V. Lochab** and F. Denef. Higher-loop vacuum integrals on S^D. arXiv:2610.30489 (2026).
2. †J. Bonifacio, K. Hinterbichler, Y. T. Albert Law, and **V. Lochab**. De Sitter observers and an effective field theory of horizon edge modes. arXiv:2610.28719v1 (2026).
3. †Y. T. Albert Law and **V. Lochab**. Horizon edge partition functions in $\Lambda > 0$ quantum gravity. *Physical Review Letters* **136**, 151601 (2026). arXiv:2603.20913.
4. †Y. T. Albert Law and **V. Lochab**. Gravitons on Nariai edges. *Physical Review D* **113**, 086006 (2026). arXiv:2506.02142.
5. X. Zheng, J. Dolde, **V. Lochab**, B. N. Merriman, H. Li, and S. Kolkowitz. Differential clock comparisons with a multiplexed optical lattice clock. *Nature* **602**, 425–430 (2022).

In preparation

6. **V. Lochab**. Three-loop melon on spheres.
7. **V. Lochab**. A finite field approach to higher-loop vacuum integrals on spheres.
8. †M. Grewal, Y. T. A. Law, and **V. Lochab**. Spinning thermal static patch correlators.
9. **V. Lochab**. A Lorentzian interpretation of the Harish-Chandra character for Dirac fields.

Fluid Dynamics at the Microscale

10. **V. Lochab**, E. D. Ewim, and S. Prakash. Continuous flow microfluidics for colloidal particle assembly on porous substrates. *Soft Matter* **19**, 2564–2569 (2023).
11. **V. Lochab** and S. Prakash. Combined electrokinetic and shear flows control colloidal particle distribution across microchannel cross-sections. *Soft Matter* **17**, 611–620 (2021).
12. **V. Lochab**, A. Yee, M. Yoda, A. T. Conlisk, and S. Prakash. Dynamics of colloidal particles in microchannels under combined pressure and electric potential gradients. *Microfluidics and Nanofluidics* **23**, 134 (2019).

Microbiology and Physiology

13. **V. Lochab**, T. H. Jones, D. H. Dusane, C. W. Peters, P. Stoodley, D. J. Wozniak, V. V. Subramaniam, and S. Prakash. Ultrastructure imaging of *Pseudomonas aeruginosa* lawn biofilms and eradication of the tobramycin-resistant variants under *in vitro* electroceutical treatment. *Scientific Reports* **10**, 9879 (2020).
14. **V. Lochab**, T. H. Jones, E. Alkandry, J. D. West, M. H. Abdel-Rahman, V. V. Subramaniam, and S. Prakash. Evaluation of electrical properties of *ex vivo* human hepatic tissue with metastatic colorectal cancer. *Physiological Measurement* **41**, 085005 (2020).
15. R. Heald, M. Bennett, V. V. Subramaniam, D. Dusane, **V. Lochab**, P. M. Sundaram, S. Salyer, J. D. West, P. Stoodley, and S. Prakash. Printed electroceutical dressings for the inhibition of biofilms and treatment of chronic wounds. *Journal of Microelectromechanical Systems* **29**, 918–923 (2020).
16. D. H. Dusane*, **V. Lochab***, T. Jones*, C. W. Peters, D. Sindeldecker, A. Das, S. Roy, C. K. Sen, V. V. Subramaniam, D. J. Wozniak, S. Prakash, and P. Stoodley. Electroceutical treatment of *Pseudomonas aeruginosa* biofilms. *Scientific Reports* **9**, 2008 (2019). **Equal Contribution*
17. S. Roy, S. Prakash, S. S. Mathew-Steiner, P. D. Ghatak, **V. Lochab**, T. H. Jones, P. M. Sundaram, G. M. Gordillo, V. V. Subramaniam, and C. K. Sen. Disposable patterned electroceutical dressing (PED-10) is safe for treatment of open clinical chronic wounds. *Advances in Wound Care* **8**, 149–159 (2019).

Conference Publications

Fluid Dynamics at the Microscale

18. **V. Lochab**, A. Yee, Y. Li, M. Yoda, A. T. Conlisk, and S. Prakash. Directed self-assembly of colloidal particles for high aspect ratio bands. *Technical Digest of the Hilton Head Workshop: A Solid-State Sensors, Actuators and Microsystems Workshop*, Hilton Head, South Carolina, pp. 265–268 (2018).
19. K. K. Rangharajan, **V. Lochab**, and S. Prakash. Desalting high salinity shale flowback water via high-flux nanofluidic evaporation-condensation. *IEEE 30th International Conference on Micro Electro Mechanical Systems (MEMS)*, Las Vegas, Nevada, pp. 60–63 (2017).
20. C. Boone, **V. Lochab**, M. Fuest, and S. Prakash. Time-dependent gating in nanofluidic channels for active electric double layer modulation for ion transport control. *Technical Digest of the Hilton Head Workshop: A Solid-State Sensors, Actuators and Microsystems Workshop*, Hilton Head, South Carolina, pp. 198–201 (2016).

Microbiology and Physiology

21. **V. Lochab**, E. Alkandry, T. H. Jones, M. Abdel-Rahman, E. W. Martin Jr., C. R. Schmidt, S. R. Abdel-Misih, M. Bloomston, V. V. Subramaniam, and S. Prakash. μ-Electrochemical impedance spectroscopy for identifying spatial heterogeneity in tumor bearing hepatic tissue. *21st International Conference on Miniaturized Systems for Chemistry and Life Sciences (μTAS)*, pp. 501–502 (2017).
22. **V. Lochab**, T. H. Jones, D. H. Dusane, C. K. Sen, S. Roy, P. Stoodley, D. Wozniak, V. V. Subramaniam, and S. Prakash. Mechanistic determination of electroceutical bactericidal effects on *Pseudomonas aeruginosa* biofilms. *21st International Conference on Miniaturized Systems for Chemistry and Life Sciences (μTAS)*, pp. 1175–1176 (2017).

Meetings & Abstracts

Physics

1. X. Zheng, J. C. Dolde, **V. Lochab**, B. N. Merriman, H. Li, and S. Kolkowitz. Measuring gravitational redshift at the centimeter scale with a multiplexed strontium optical lattice clock. *Bulletin of the American Physical Society* **66**, V01.00133 (2021).
2. J. C. Dolde, X. Zheng, **V. Lochab**, B. N. Merriman, H. Li, and S. Kolkowitz. Enhanced differential stability measurements with a multiplexed strontium optical lattice clock. *Bulletin of the American Physical Society* **66**, S03.00002 (2021).

Fluid Dynamics at the Microscale

3. S. Prakash, **V. Lochab**, and E. Ewim. Use of combined electrokinetic and Poiseuille flows to generate organized colloidal structures. *Bulletin of the American Physical Society* **68**, N18.00011 (2023).
4. S. Prakash, **V. Lochab**, and E. Ewim. Demonstrating use of continuous flow microfluidics to assemble colloidal particles on porous substrates. *Bulletin of the American Physical Society* **67**, U34.00007 (2022).
5. S. Prakash and **V. Lochab**. Colloidal migration in microchannels under combined potential and pressure gradients. *Bulletin of the American Physical Society* **65**, D15.00012 (2020).
6. S. Prakash and **V. Lochab**. Colloidal particle migration within microchannels in combined Poiseuille and electrokinetic flows. *Bulletin of the American Physical Society* **64**, H36.00002 (2019).
7. S. Prakash and **V. Lochab**. Electrohydrodynamic printing of pre-assembled colloidal particle structures. *Bulletin of the American Physical Society* **63**, G24.00003 (2018).
8. M. Yoda, A. Yee, **V. Lochab**, and S. Prakash. Colloidal band assembly in different microchannels. *Bulletin of the American Physical Society* **62**, G37.00002 (2017).

Fellowships & Awards

1. *Pfister Fellowship*, Columbia University, Spring 2024.
2. *Charles Townes Fellowship*, Columbia University, Fall 2021.
3. *Future Academic Scholars Training (FAST) Fellowship*, The Ohio State University, Fall 2017 – Spring 2018.
4. *Center for Energy Research, Training, and Innovation (CERTAIN) Travel Award*, The Ohio State University, 2018.
5. *Student Author Travel Award*, A Solid-State Sensors, Actuators and Microsystems Workshop, The Ohio State University, 2018.
6. Engine and design team member of the Formula Student car team at IIT (ISM) Dhanbad that placed third in design at SUPRA SAEINDIA 2012.
7. Engine and design team member of the Formula Student car team at IIT (ISM) Dhanbad that placed first in the autocross event for maneuverability at SUPRA SAEINDIA 2011.

Teaching

Columbia University — Graduate Teaching Assistant
Quantum Mechanics II (G); Quantum Field Theory I (G); Electromagnetic Theory (G); Thermal and Statistical Physics (UG); Mathematical Methods of Physics (UG); General Physics I and II Laboratories (UG). G: graduate; UG: undergraduate.

The Ohio State University — Graduate Teaching Assistant
Measurement Systems and Data Analysis (UG).

Technical Skills

1. Python, C++, MATLAB, Mathematica, COMSOL Multiphysics, AutoCAD, CATIA.
2. Finite-difference methods; basic molecular dynamics simulations using Nanoscale Molecular Dynamics; Monte Carlo methods.
3. Five years of cleanroom experience in micro- and nanofabrication for microfluidic applications, with training in semiconductor device fabrication techniques including photolithography, oxidation, diffusion, chemical processes, reactive ion etching, and metallization.
4. Imaging and spectroscopy experience: Transmission Electron Microscopy, Scanning Electron Microscopy, Atomic Force Microscopy, Confocal Fluorescence Microscopy, Raman Spectroscopy, Fourier-transform Infrared Spectroscopy (FTIR) and Microscopy.
5. Engine testing and development (1.5 years), including motorcycle powertrain testing on automotive test beds and analysis of component performance and efficiency.
6. Microbiology lab experience (3 years) in a biosafety level 2 (BSL-2) facility.